

Week One - Proposal

1. Team Information

Team name: Team MAI

Members:

Damirova Assem - 23B030815 - EDA Lead

Murat Madiyar - 24B031904 - Modelling Lead

Askarkbekova Ingkar - 24B031655 - Report Lead & Presentation Lead

2. Dataset Selection

Dataset Name: Student Performance

Source URL: <https://archive.ics.uci.edu/dataset/320>

Dataset Size: 649 rows and 30 columns

Confirmation: The dataset has been downloaded and successfully opened in Python.

3. Problem Type

Problem Type: Regression

Target Variable: Final grade (G3)

Goal: The goal of this project is to predict students' final academic performance (G3) based on demographic, social, and academic factors.

4. Hypothesis

We hypothesize that a combination of academic (G1, G2), behavioral (study time, absences), and social factors are the strongest predictors of the final grade (G3).

5. Planned Feature Approach

Previous grades (G1, G2): Strong indicators of student performance trends and likely predictors of final grade.

Social factors: Social features such as family support, parental education, and lifestyle habits are expected to impact student performance.

Absences: High number of absences is expected to negatively affect academic performance due to missed learning time.

- Categorical features such as school, sex, and family support will require encoding (e.g., one-hot encoding) before modelling.
- This dataset was selected because it provides a realistic regression problem with multiple influencing factors.